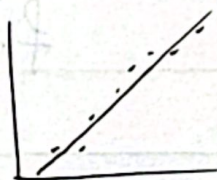


0427

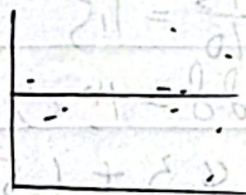
5. Normal Probability Plot

⇒ 殘差呈常態分配



Versus Fits

⇒ 喇叭狀，誤差的變異數隨預測值的增大而增大，不符合殘差變異數的一致性的假設



8a. Linearity: $E(\epsilon) = 0$. Independence: 誤差項之間相互獨立

Normality: ϵ 服從常態分配, Equal Variance: 對於所有 x 值誤差項的變異數都是相同的常數

b. σ^2 的不偏估計值為 $MS_e = 58.05$

c. Normal Probability Plot: 紅點貼近藍色的直線
⇒ 殘差分配接近常態

Versus Fits: 點呈隨機分布，沒有明顯規律，符合變異數-一致性

Summary: 此迴歸模型前題假設有效且合理

9a. $r^2 = 0.027 = 2.7\%$ ⇒ 價格和螢幕尺寸之間的線性關係非常微弱

b. 常態性、變異數-一致性沒有被違反

c. Linearity 被違反，價格並不隨尺寸呈簡單線性遞增

$$17. \bar{x} = \frac{2 \times 5 + 3 \times 5 + 4 \times 5}{15} = 3$$

$$\bar{y} = 8.8$$

$$S_{xx} = 10$$

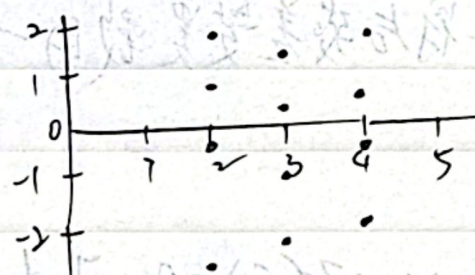
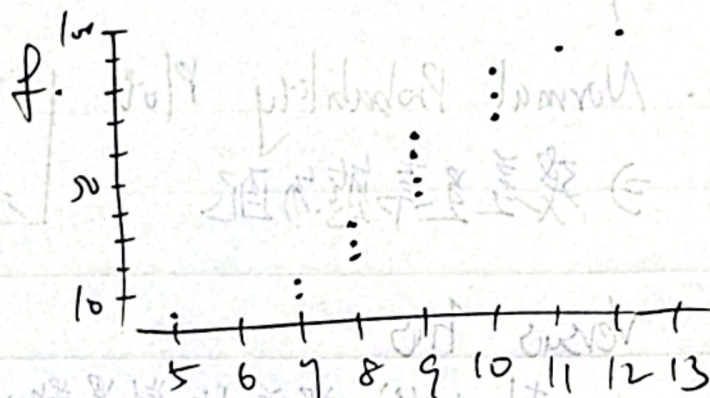
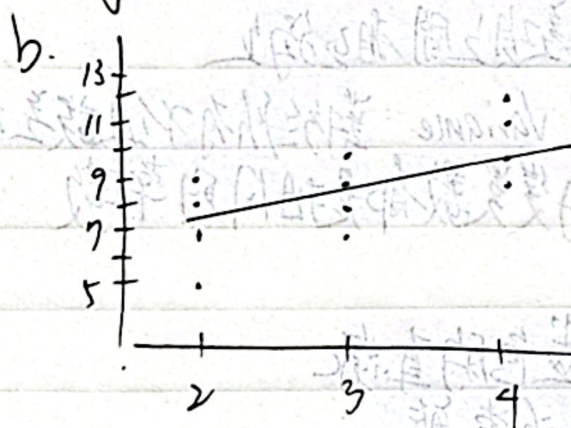
$$S_{yy} = 42.4$$

$$S_{xy} = 15$$

$$a. b = \frac{15}{10} = 1.5$$

$$a = 8.8 - 1.5 \times 3 = 4.3$$

$$\hat{y} = 4.3 + 1.5x$$



符合常態性與變異數一致性

$$c. SSE = TSS - SSR = 42.4 - \frac{15^2}{15} = 19.9$$

$$MSE = 19.9 \div 13 \approx 1.5308$$

$$d. H_0: \beta = 0, H_a: \beta \neq 0$$

$$t_{STAT} = \frac{1.5}{\sqrt{\frac{1.5308}{10}}} \approx 3.83 > t_{13, 0.005} = 2.160 \Rightarrow \text{Reject } H_0$$

⇒ 兩者存在顯著的線性關係

$$e. p\text{-value} = 0.0027 < 0.05 \Rightarrow \text{Reject } H_0. \text{ 結論同 (d)}$$

$$g. \hat{y} = 1.5 \times 3 + 4.3 = 8.8$$

$$95\% \text{ C.I.} = b \pm t_{13, 0.005} \times \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

$$= 8.8 \pm 2.160 \times \sqrt{1.53 \left(\frac{1}{15} + \frac{(3-3)^2}{10} \right)} = [8.1, 9.49]$$

$$15. a. \hat{y} = 2 - 0.875x, \bar{x} = 0, \bar{y} = 2$$

$$MSE = 0.0833, S_{xx} = 16$$

$$\begin{aligned} 99\% \text{ C.I.} &= \hat{y} \pm t_{3, 0.005} \times \sqrt{MSE \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \\ &= 2.875 \pm 5.841 \times \sqrt{0.0833 \left(\frac{1}{5} + \frac{(-1)^2}{16} \right)} \\ &= [2.011, 3.739] \end{aligned}$$

$$\begin{aligned} b. 99\% \text{ P.I.} &= \hat{y} \pm t_{3, 0.005} \times \sqrt{MSE \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \\ &= 1.125 \pm 5.841 \times \sqrt{0.0833 \left(1 + \frac{1}{5} + \frac{1^2}{16} \right)} \\ &= [-0.769, 3.019] \end{aligned}$$

c. 當 $x_0 = \bar{x}$ 時，常預測值 $x_0 = \bar{x} = 0$ ，PI 會最小。